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International Autumn School

CP2K-Newton-X-Libra-WaveMixings
for mixed quantum-classical dynamics

October 14-16, 2026, Kiel, Germany

Book of Abstracts



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S U R F A C E &
I N T E R F A C E
S C I E N C E

Christian-Albrechts-University Kiel, Germany

Schedule for the Autumn School

Day 1 (Oct 14)

13:30 pm	Welcome (Anna)
13:45 - 14:45 pm	Lecture (Anna)
14:45 - 15:45 pm	Lecture (Philipp)
Break	
16:00 - 16:30 pm	HPC Systems at CAU
16:30 - 18:00 pm	Hands-on session (Anna, Tejas)

Day 2 (Oct 15)

09:00 am	Lecture (Mario)
10:00 - 10:30 am	Lightning talks participants
10:30-12:00 am	Hands-on session (Ritama, Rafael)
Lunch	
14:00 - 15:30 pm	Poster session
15:30 - 16:30 pm	Lecture (Stepan)
Break	
17:00 - 18:30 pm	Lecture / Hands-on session (Alexey / Mohammad)
19:00	Social Dinner (self-paid)

Day 3 (Oct 16)

9:00 - 10:30 am	Hands-on session (Ritama, Rafael)
10:30 -12:00 am	Lecture / Hands-on session (Maxim / Luis)
Lunch	
14:00 - 15:00 pm	Lecture (Ole)
15:00 - 16:00 pm	Lecture (Anna)
16:00 pm	Closing (Anna)

Lightning Talks

Day 2 (Oct 15)

- 10:00 am** **Vinicius do Nascimento da Rocha**
"Design Principles for Next Generation of Small Organic Molecules for Photodynamics Therapy Revealed by Nonadiabatic Molecular Dynamics"
- 10:05 am** **Dario Calvani**
"From Metal Poly(Heptazine Imide) Active Sites to CO₂ Photoreduction Catalysis: A Multiscale Electronic Structure Perspective"
- 10:10 am** **Michael Ingham**
"Trajectory surface hopping in inorganic photocatalysts: Application to TiO₂ and BiVO₄ nanoclusters and bulk crystal"
- 10:15 am** **Gabriel Rodrigues**
"Variational MC-PDFT including Range-Separation: Strong Correlation? Not so Strong Anymore"
- 10:20 am** **Andrea Carolina Chimarro Contreras**
"How to unveil the Excited-State Photoreactivity of oxime esters?"

Design Principles for Next Generation of Small Organic Molecules for Photodynamics Therapy Revealed by Nonadiabatic Molecular Dynamics

Vinícius do Nascimento da Rocha¹, Paulo Cesar Piquini¹, Davide Avagliano²

¹Universidade Federal de Santa Maria (*Physics*) , ²École Nationale Supérieure de Chimie de Paris (*Chemistry*)

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Nitrobenzochalcogenadiazole derivatives are emerging candidates for Photodynamic Therapy (PDT), yet the precise mechanisms governing their excited-state deactivation and triplet generation remain to be fully elucidated. This study employs non-adiabatic dynamics simulations, using the Trajectory Surface Hopping method within a Linear Vibronic Coupling (LVC) framework, to unravel the Intersystem Crossing (ISC) pathways in these systems. We systematically investigate two design factors: the heavy-atom effect (substituting S with Se and Te) and the influence of “push-pull” electronic architectures (Donor-Acceptor vs. Donor-Donor/Acceptor-Acceptor). Our results demonstrate that replacing Sulfur with Selenium and Tellurium monotonically accelerates ISC, reducing excited-state lifetimes from ≈ 6.1 ps to sub-picosecond timescales (≈ 0.9 ps) via a dominant $S_2 \rightarrow S_1 \rightarrow T_n$ relaxation channel driven by enhanced spin-orbit coupling. Furthermore, we reveal that structural modifications that disrupt this push-pull nature (D-D or A-A) result in kinetic bottlenecks, trapping the population in the singlet manifold. These dynamical insights establish clear structure-property relationships, guiding the rational design of photosensitizers with optimized triplet quantum yields.

Acknowledgements

This study was financed by Conselho Nacional de Desenvolvimento Científico e Tecnológico - CNPq (Grants PQ # 311161/2022-7 and Universal #420820/2025-6), and the Fundação de Amparo à Pesquisa do Estado do Rio Grande do Sul - FAPERGS (Project # 24/2551-0001290-7)

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ROCHA, Vinícius N. da; AVAGLIANO, Davide; PIQUINI, Paulo C. Design principles for next generation of small organic molecules for photodynamics therapy revealed by nonadiabatic molecular dynamics. *Chemistry - A European Journal*, 2026. DOI: <https://doi.org/10.1002/chem.70941>

From Metal Poly(Heptazine Imide) Active Sites to CO₂ Photoreduction Catalysis: A Multiscale Electronic Structure Perspective

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Abstract:

Metal poly(heptazine imide) (M-PHI), a two-dimensional crystalline carbon nitride, is a promising platform for single-atom and surface catalysis, although its active site structures remain unresolved.^(1,2) Here, we elucidate the catalytic site structure of nickel poly(heptazine imide) (Ni-PHI) using a combined experimental–computational approach integrating synthesis, CO-DRIFTS spectroscopy, and density functional theory (DFT)-based static and molecular dynamics simulations with vibrational density-of-states analysis. By correlating experimental CO stretching features with simulated spectra across different nickel loadings and aggregation states, we identify the most probable Ni active site configurations, capturing both single-atom and nanoparticle scenarios. This establishes a robust framework for atomistic characterization of catalytic sites in crystalline carbon nitrides.⁽³⁾ Building on this, we propose a future ab initio investigation of photoinduced CO₂ reduction on M-PHI,⁽⁴⁾ aimed at resolving excited-state charge transfer, CO₂ activation mechanisms, and photodynamics–activity relationships under photocatalytic conditions.

References:

- (1) Zhou, M.; Ou, H.; Liu, Z.; Jiang, Z.; Yang, C.; Fang, Y.; Wang, S.; Zhang, G.; Cheng, J.; Hou, Y.; Wang, X. Poly(Heptazine Imide): Crystalline Allotrope of Polymeric Carbon Nitrides for Solar to Chemical Energy Conversion. *Angew Chem Int Ed* 2026, 65 (14), e1708637. <https://doi.org/10.1002/anie.1708637>.
- (2) Pelicano, C. M.; Antonietti, M. Metal Poly(Heptazine Imides) as Multifunctional Photocatalysts for Solar Fuel Production. *Angew Chem Int Ed* 2024, 63 (24), e202406290. <https://doi.org/10.1002/anie.202406290>.
- (3) Calvani, D.; Farias, I. G.; Barreto, D.N.; Noleto, L. F. G.; da Silva, M. A. R.; Teixeira, I.; Kuc, A. B.; Kühne, T. D. Understanding the Catalytic Site Structure in Metal Poly(Heptazine Imide) – Crystalline Carbon Nitride. *J. Mater. Chem. A*, **2026**. <https://doi.org/10.1039/D6TA01987K>.
- (4) Pios, S.; Domcke, W. Ab Initio Electronic Structure Study of the Photoinduced Reduction of Carbon Dioxide with the Heptazinyl Radical. *J. Phys. Chem. A* **2022**, 126 (18), 2778–2787. <https://doi.org/10.1021/acs.jpca.2c00615>.

Trajectory surface hopping in inorganic photocatalysts: Application to TiO_2 and BiVO_4 nanoclusters and bulk crystal

Michael Ingham[†] and Rachel Crespo-Otero[†]

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Photocatalytic water-splitting is a promising route for efficient solar energy conversion, where TiO_2 is the prototypical material due to its low cost, relative abundance, and non-toxicity. Nonadiabatic dynamics simulations, such as Trajectory Surface Hopping (TSH), provide a promising route to disentangling the web of competing ultrafast radiative and nonradiative processes that occur following light absorption. For example, TDDFT-based TSH simulations have been utilised to elucidate the water-splitting mechanism in a TiO_2 nanocluster.¹ Compared to molecules, TSH in bulk solids remains challenging due to the complexity of the crystal potential energy surfaces, the large number of states required to fully capture the excitonic profile, and the associated cost of such calculations in plane wave codes. A recent interface between the CP2K² package, for highly-efficient periodic Tamm-Dancoff Approximation (TDA) calculations in the Gaussians and Plane Waves (GPW) method, and **Newton-X**,³ for efficient TSH and determination of approximate nonadiabatic couplings, provides an effective route to efficient TSH in bulk solids. As such, the objective of the present work is to explore the feasibility of such approach in inorganic materials. We study TiO_2 as a test case, which is compared to another promising photocatalyst, BiVO_4 .⁴ Both a comprehensive characterisation of the optical properties of a set of nanoclusters, hydrated nanoclusters is performed, as well as TSH simulations to see the excited states evolve dynamically. Cluster and periodic models are compared on a level footing, providing a unique bridge between localised and delocalised approaches to materials modelling.

References

- [1] M. Muuronen, S. M. Parker, E. Berardo, A. Le, M. A. Zwijnenburg and F. Furche, *Chemical Science*, 2017, **8**, 2179–2183.
- [2] T. D. Kühne, M. Iannuzzi, M. Del Ben, V. V. Rybkin, P. Seewald, F. Stein, T. Laino, R. Z. Khaliullin, O. Schütt, F. Schiffmann, D. Golze, J. Wilhelm, S. Chulkov, M. H. Bani-Hashemian, V. Weber, U. Borštnik, M. Taillefumier, A. S. Jakobovits, A. Lazzaro, H. Pabst, T. Müller, R. Schade, M. Guidon, S. Andermatt, N. Holmberg, G. K. Schenter, A. Hehn, A. Bussy, F. Belleflamme, G. Tabacchi, A. Glöß, M. Lass, I. Bethune, C. J. Mundy, C. Plessl, M. Watkins, J. VandeVondele, M. Krack and J. Hutter, *The Journal of Chemical Physics*, 2020, **152**, 194103.
- [3] M. Barbatti, R. S. Mattos, B. Demoulin, M. d. O. Bispo, M. Bondanza, M. Brady, R. Crespo-Otero, E. G. F. d. Miranda, P. O. Dral, G. Granucci, A. Hehn, F. J. Hernández, G. Iuzzolino, R. Kar, F. Kossoski, H. Lischka, B. Mennucci, S. Mukherjee, A. Mukhopadhyay, F. Perrella, M. Persico, M. Pinheiro, J. Pittner, F. Plasser, N. Rega, E. Sangiogo-Gil, T. Thorat, J. M. Toldo, A. A. Tomaz, M. T. d. N. Varela and L. Vasquez, *Physical Chemistry Chemical Physics*, 2026.

Variational MC-PDFT including Range-Separation: Strong Correlation? Not so Strong Anymore

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Strongly correlated systems remain a challenge in modern computational chemistry, as DFT often fails to describe them quantitatively, and more adequate methods are too expensive for most practical purposes. However, the proper treatment of strong correlation is extremely important because it frequently appears in ordinary cases such as metallic complexes and excited states. It is therefore critical to develop accurate, but less costly methods. For this, multiconfigurational pair-density functional theory (MC-PDFT) has shown significant potential due to its proper treatment of static correlation while efficiently incorporating dynamic correlation through the pair-density potential. However, most existing implementations compute the PDFT energy in a post-SCF manner, thus ignoring the orbital relaxation effects induced by the pair-density exchange-correlation functional.

In this work, we present recent investigations [1,2] of the performance of our new variational implementation [3] of MC-PDFT developed in the MultiPsi software [4]. It includes translated functionals of GGA and hybrid-GGA form, as well as a new scheme for range separation (MC-sr-PDFT) [5], both for state-specific (SS) and state-averaged (SA) wavefunctions.

We showcase results of variational SA-MC-(sr-)PDFT for excited states of small open-shell molecules and organic chromophores, and, using the state-specific formalism, the singlet-triplet gaps of large conjugated hydrocarbons with potential for quantum applications. Our results already show that when range separation is included, MC-sr-PDFT is comparable to NEVPT4, being (for small radicals) remarkably superior to regular (broken-symmetry) DFT or EOM-CCSD, with mean absolute errors against MRCISD+Q as low as 0.06 eV.

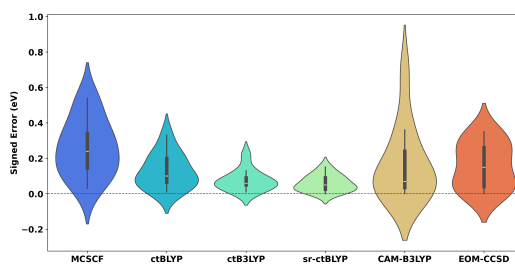


Figure 1: Violin plots of distributed signed errors of different methods against MRCISD+Q. PDFT translated-functionals start with “ct”, where the inclusion of “sr-” indicates a range-separation scheme.

References:

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How to unveil the Excited-State Photoreactivity of oxime esters?

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The chemistry of a molecular specie in excited states can be significantly different from its chemistry in the ground state. Excess of energy of an excited specie can help to overcome activation barriers and, as different states can be populated, new reaction pathways can be opened.^[1] Several computational methods have been used to model excited state processes. TD-DFT and Δ SCF are among the most used and simpler approaches to perform static studies of a system. This involves, for example, computing excitation energies and mapping excited states potential energy surfaces along selected coordinates.^[2] Even though this approach gives us valuable information, such as energies profiles and activation barriers, it relies on prior knowledge of the reaction coordinate and/or chemical intuition. Due to this limitation, non-adiabatic molecular dynamics (NAMD) approaches are extremely helpful to explore excited states landscape. These methods can explore different excited states pathways and follows the evolution of a molecular geometry towards one or several photoproducts upon light absorption. This makes NAMD a very useful tool when different competitive excited state decay pathways can play a role. Furthermore, these approaches can give information on the excited state lifetimes. However, as they combine nuclear classical dynamics with quantum chemical description, they can be very computationally demanding, being strongly depending on the electronic structure level needed for the investigation.^[3]

In this work, we modelled the photo induced dissociation of 4 oxime esters^[4] (Figure 1) using static and dynamic approaches. First, we simulated the UV-Vis spectra of the oxime esters, followed by the calculation of potential energy surfaces along the different reaction coordinates. After verifying that the electronic structure method was able to describe the photoreaction of interest, we performed NAMD simulations using Surface Hopping. The dynamics reveal that a twist around a dihedral angle θ is necessary to prior to proceed with the photodissociation and it is noteworthy that photoisomerization and photodissociation are competitive deactivation pathways. This work shows how computational chemistry can be used to unveil complicated excited states chemistry.

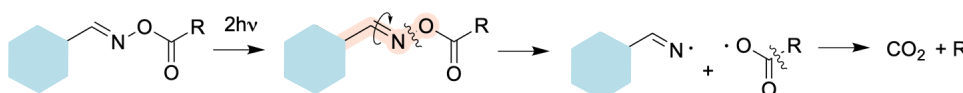


Figure 1. Proposed photocleavage mechanism of oxime esters. Dihedral angle θ is highlighted.

Acknowledgements. This work has been supported by ANR-25-CE29-2684-01 (SupraPhoMo) and ENS de Lyon.

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- [2] Herbert, J., *Phys. Chem. Chem. Phys.* **2024**, 26, 3755-3794. <https://doi.org/10.1039/D3CP04226J>
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Posters

Day 2 (Oct 15), 14:00 - 15:30 pm

Dario Calvani

"From Metal Poly(Heptazine Imide) Active Sites to CO₂ Photoreduction Catalysis: A Multiscale Electronic Structure Perspective"

Michael Ingham

"Trajectory surface hopping in inorganic photocatalysts: Application to TiO₂ and BiVO₄ nanoclusters and bulk crystal"

Björn Borgolte

"Multi-scale fragment simulation of molecular photoswitches"

Mateus Bergami

"Towards Predictive QM/MM Models of Solution-Processed Organic Semiconductors"

Lenard Carroll

"TBA"

Singaravel Nathiya

"A Theoretical Study of Heavy Atom and Oxidation Effects in MR-TADF Emitters for OLEDs: DFT, Double-Hybrid DFT, CCSD, & QM/MM Approaches"

Masiyappan Karuppusamy

"Topology-Dependent Adsorption, Rotational Stability, and Work-Function Modulation of Polycyclic Aromatic Hydrocarbons on Au(111): A Density Functional Theory Study"

Krishnandu Dey

"TBA"

From Metal Poly(Heptazine Imide) Active Sites to CO₂ Photoreduction Catalysis: A Multiscale Electronic Structure Perspective

Dario Calvani,^{1,2} Agnieszka Kuc,^{1,2} Thomas D. Kühne.^{1,2}

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Abstract:

Metal poly(heptazine imide) (M-PHI), a two-dimensional crystalline carbon nitride, is a promising platform for single-atom and surface catalysis, although its active site structures remain unresolved.^(1,2) Here, we elucidate the catalytic site structure of nickel poly(heptazine imide) (Ni-PHI) using a combined experimental-computational approach integrating synthesis, CO-DRIFTS spectroscopy, and density functional theory (DFT)-based static and molecular dynamics simulations with vibrational density-of-states analysis. By correlating experimental CO stretching features with simulated spectra across different nickel loadings and aggregation states, we identify the most probable Ni active site configurations, capturing both single-atom and nanoparticle scenarios. This establishes a robust framework for atomistic characterization of catalytic sites in crystalline carbon nitrides.⁽³⁾ Building on this, we propose a future ab initio investigation of photoinduced CO₂ reduction on M-PHI,⁽⁴⁾ aimed at resolving excited-state charge transfer, CO₂ activation mechanisms, and photodynamics-activity relationships under photocatalytic conditions.

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Trajectory surface hopping in inorganic photocatalysts: Application to TiO_2 and BiVO_4 nanoclusters and bulk crystal

Michael Ingham[†] and Rachel Crespo-Otero[†]

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References

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- [3] M. Barbatti, R. S. Mattos, B. Demoulin, M. d. O. Bispo, M. Bondanza, M. Brady, R. Crespo-Otero, E. G. F. d. Miranda, P. O. Dral, G. Granucci, A. Hehn, F. J. Hernández, G. Iuzzolino, R. Kar, F. Kossoski, H. Lischka, B. Mennucci, S. Mukherjee, A. Mukhopadhyay, F. Perrella, M. Persico, M. Pinheiro, J. Pittner, F. Plasser, N. Rega, E. Sangiogo-Gil, T. Thorat, J. M. Toldo, A. A. Tomaz, M. T. d. N. Varella and L. Vasquez, *Physical Chemistry Chemical Physics*, 2026.

Multi-scale fragment simulation of molecular photoswitches

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Some chemical compounds exhibit reversible isomerization upon irradiation with light, allowing for a controlled change of their physical and chemical properties. Recently, these molecular photoswitches have shown increasing potential as key components in nanotechnology ranging from smart materials to pharmacology and information storage [1]. The understanding of the quantum mechanical mechanisms behind isolated photoswitches has advanced considerably with widely available numerical frameworks for nonadiabatic molecular dynamics and Quantum Mechanics/Molecular Mechanics approaches [2]. Such simulations of photoswitches nevertheless face difficulties due to the combination of the excited state electronic structure and highly nonadiabatic core dynamics. The description of interacting photoswitchable structures, for example as part of a spatially extended membrane or a biological macromolecule, exasperates these issues. Generally applicable numerical frameworks that can adequately simulate the dynamics of such structures with photoswitchable active centers are part of ongoing research [3]. One approach towards numerical accessibility is the adaptation of existing multiscale-multilayer fragment methods, which are tried-and-tested models for many-atom interactions, to the required excited regime. With the project still in its beginning phase, the scope and goal of such adaptation is presented. The problems with numerical treatment of the dynamics of photoswitches, such as the breakdown of the Born-Oppenheimer approximation and scale differences, are discussed. Particular focus is placed on the resolution of these issues by adopting multi-scale fragment methods.

- [1] Zbigniew L. Pianowski. “Recent Implementations of Molecular Photoswitches into Smart Materials and Biological Systems”. In: *Chemistry – A European Journal* 25.20 (2019), pp. 5128–5144. DOI: <https://doi.org/10.1002/chem.201805814>.
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- [3] Amirhossein Bakhtiari et al. “A Multiscale Simulation Framework for Elucidating Photochemical Structure–Activity Relationships of Photoswitchable Ligands in Complex Biomolecular Environments”. In: *Journal of Chemical Information and Modeling* 65.23 (2025), pp. 12861–12878. DOI: 10.1021/acs.jcim.5c01831.

Title: Towards Predictive QM/MM Models of Solution-Processed Organic Semiconductors

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¹*Department of Engineering and Physics, Karlstad University, Sweden.*

²*Laboratory for Chemistry of Novel Materials, University of Mons, Belgium.*

³*Namur Institute of Structured Matter, University of Namur, Belgium.*

⁴*Materials Theory Division, Department of Physics and Astronomy, Uppsala University, Sweden.*

Predictive multiscale models capable of describing condensed-phase environments are essential for understanding the electronic processes governing the performance of organic semiconductors in energy-conversion applications. In this contribution, we present a multiscale QM/MM framework designed to investigate how different descriptions of the environment influence the electronic properties governing charge generation and transport. Particular attention is devoted to the comparison between continuum approaches, represented by the polarizable continuum model (PCM), and atomistic embedding strategies, including electrostatic and polarizable embedding. These complementary models enable a systematic assessment of the balance between computational efficiency and physical realism in describing complex condensed-phase environments. The methodology is applied to representative benchmark systems, including Y6 neat films and PM6 bulk heterojunctions, to investigate properties relevant to device performance, such as redox potentials, singlet and triplet excited states, UV-Vis absorption spectra, exciton characteristics, and electron and hole transfer processes. By establishing a consistent framework for comparing continuum and atomistic descriptions, this work provides insights into the role of environmental effects in determining the electronic structure of organic semiconductors and offers practical guidelines for selecting appropriate embedding strategies in large-scale simulations of functional energy materials.

Lenard Carroll

The intercalation of halogen molecules into 2D graphitic systems (such as graphite, bilayer graphene and carbon nanotubes) offers a powerful strategy for tuning their electronic, optical, and structural properties. Halogen intercalants are particularly effective in this regard, as even small charge transfers (towards the intercalants) have been shown to significantly modify the electronic structure of the host material, making halogen-intercalated graphite intercalation compounds (GICs) attractive candidates for tunable electronic applications. While bromine-based stage-2 GICs have been extensively studied, with the formation of extended polybromide networks well-documented, chlorine-based GICs remain comparatively underexplored (due to experimental challenges), leaving open questions about their structural and electronic behavior.

To address this gap, we employ first-principles calculations in the form of density functional theory (DFT) and *ab initio* molecular dynamics (AIMD), to investigate the formation of polyhalide networks (using Br₂, Cl₂ and BrCl) in AB-stacked graphite and packed carbon nanotubes, while using Bader charge and density of states analyses to gain insight into their electronic structure. Preliminary DFT results from AG Paulus reveal that all studied configurations of Cl₂ intercalation in AB-stacked graphite are thermodynamically stable, with intercalation energies that depend strongly on the resulting polychloride structure. Depending on concentration and supercell size, distinct structural motifs emerge, including one-dimensional chains. These configurations exhibit significant Cl-Cl bond elongation and a charge transfer of approximately 0.01 electrons per carbon atom to the intercalants. Building on these findings, this work aims to deepen our understanding of the charge transfer mechanisms governing polyhalide chain formation across graphitic host materials.

A Theoretical Study of Heavy Atom and Oxidation Effects in MR-TADF Emitters for OLEDs: DFT, Double-Hybrid DFT, CCSD, & QM/MM Approaches

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Highlights

This study analyzes 48 MR-TADF emitters, incorporating heavy atoms, oxidation, and donor effects. DFT, TD-DFT, and QM/MM reveal enhanced SOC, reduced ΔE_{ST} , and improved k_{ISC} ($\sim 10^8$ s⁻¹) and k_{RISC} ($\sim 10^6$ s⁻¹). Eight potential molecules were further studied using advanced DFT and CCSD methods, highlighting their potential for high-efficiency OLEDs.

Abstract

The emerging multi-resonance thermally activated delayed fluorescence (MR-TADF) emitters with organoboron and nitrogen cores highlight their significance in OLEDs.¹ However, their efficiency is challenged by slower rate constants in the reverse intersystem crossing (k_{RISC}) process compared to conventional TADF emitters. The study entails an in-depth analysis focused on gaining a better understanding of the photophysical properties of MR-TADF emitters. Using DFT and TD-DFT analyses, 48 MR-TADF molecules are studied, incorporating heavy atoms such as sulfur and selenium, and their subsequent oxidation, and peripheral donors such as carbazole (Cz), *tert*-butyl-carbazole (tCz), diphenylacridine (DPAC), and dimethylacridine (DMAC) into organo boron and nitrogen-embedded systems. Moreover, the QM/MM approach was utilized to examine the excited state properties in crystal phase. A comprehensive assessment of this molecular framework reveals that integrating heavy atoms and donors into MR-TADF molecules results in significant enhancements in ΔE_{ST} , larger SOC, and higher-order radiative (10^{+08} s⁻¹) rates, leading to faster k_{ISC} ($\sim 10^{+08}$ s⁻¹) and k_{RISC} ($\sim 10^{+06}$ s⁻¹) rates. Based on key criteria, eight potential molecules were selected and to precisely analyze the excited state properties using double-hybrid density functionals² including B2PLYP and PBE0-2, along with highly correlated wave function STEOM-DLPNO-CCSD.³ All DFT and TD-DFT calculations are performed using Gaussian 09 Software.⁴ All SOC and double-hybrid density functionals⁵ are carried out using ORCA-5.0 Software.

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Topology-Dependent Adsorption, Rotational Stability, and Work-Function Modulation of Polycyclic Aromatic Hydrocarbons on Au(111): A Density Functional Theory Study

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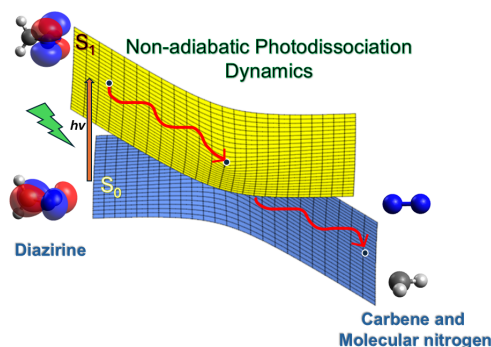
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Abstract

The adsorption behavior of polycyclic aromatic hydrocarbons (PAHs) on metal surfaces plays a critical role in determining the structural and electronic properties of organic-metal interfaces relevant to molecular electronics, optoelectronics, and surface engineering. In this work, dispersion-corrected density functional theory (DFT) calculations, as implemented in the **CP2K** package, were employed to investigate the adsorption characteristics of linear and circular PAHs on the Au(111) surface. The investigated systems include linear acenes ranging from benzene to pentacene and circular PAHs ranging from phenanthrene to dodecabenzocoronene. The calculated adsorption energies reveal that the adsorption strength generally increases with molecular size and π -conjugation, ranging from -0.765 eV for benzene to -3.520 eV for pentacene, while reaching -3.379 eV for dodecabenzocoronene. Charge-density-difference analysis indicates localized electron redistribution at the PAH/Au(111) interface while preserving the predominantly physisorptive nature of the interaction. Rotational energy calculations demonstrate that molecular topology and symmetry strongly influence orientational stability. Among the investigated molecules, pentacene exhibits the largest rotational barrier (0.111 eV), whereas coronene and tribenzocoronene display nearly barrier-free rotational behavior. Furthermore, work-function calculations reveal a systematic reduction of the Au(111) work function from 5.17 eV for the pristine surface to 4.70 eV for dodecabenzocoronene-covered Au(111), reflecting enhanced interfacial polarization with increasing π -conjugation. Overall, the results establish clear relationships between molecular topology, adsorption energetics, rotational mobility, and electronic perturbations at PAH/Au(111) interfaces. These findings provide fundamental insights into molecule-surface interactions and offer valuable guidelines for the design of functional organic-metal interfaces. The study also demonstrates CP2K's capability to investigate complex organic-metal interfaces and lays the foundation for future studies of charge transfer and excited-state phenomena at molecular surfaces.

Abstract



This work is done to understand the mechanistic details of the excited state non-adiabatic process of diazirine and three other derivatives that are substituted with electron-donating (methoxy-diazirine) and withdrawing (fluoro-diazirine) groups, as well as combining both (fluoro-methoxy-diazirine), to study their effect on generating carbene and molecular nitrogen via diazomethane. All electronic structure calculations are performed at SA2-CASSCF followed by MS-CASPT2 levels of theory using OpenMolcas, and non-adiabatic molecular dynamics simulations at SA2-CASSCF level by choosing appropriate active spaces using NewtonX-CS. Two conical intersections (CIs) are found for each of the molecules in the non-adiabatic process, which are associated with asymmetric vibrations of the C–N bonds in the three-membered ring of the diazirines, except for pure diazirine, where the second CI is associated with symmetric vibration. The dynamics starts from $n\text{-}\pi^*$ excited state (S_1). In the dynamics, trajectories are seen to go through both the CIs whichever is energetically and dynamically permissible. The first-order kinetics of N_2 formation revealed that the reaction is fastest for pure diazirine and slowest for fluoro-methoxy-diazirine. The mechanism, as revealed from the present calculation, looks so reliable that an MS-CASPT2 dynamics, if performed, would perhaps change the population in each pathway, leading to different rate constants of N_2 formation, but the number of pathways and the trend of rate constants may remain the same.